

Global Retail Business Intelligence: Multi-dimensional Data-driven Decision Support

Course: AIDM7400

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1. Introduction & Research Question

- **Research Question:**

How can a multi-dimensional analysis of global retail data—integrating temporal, logistical, product, and customer insights—drive strategic decision-making to optimize profitability and operational efficiency?

- **Project Objectives:**

The primary objective is to transform raw transaction data into actionable business intelligence. We aim to locate geographic "profit black holes" where sales are high but margins are low. The project also focuses on optimizing logistics by identifying cost-effective shipping modes. Furthermore, we verify the 80/20 Pareto principle in product performance and apply RFM modeling to segment customers for targeted retention.

2. Data Source & Preprocessing

2.1 Dataset Selection

This study uses the Global Retail Sales Dataset from Kaggle. This dataset contains transaction-level records from a multinational retail company, covering comprehensive business information including order details, customer attributes, product categories, geographic distribution, and financial metrics.

The original dataset contains 51,290 transaction records and 21 raw features. Key features cover multiple dimensions required for retail analysis:

- Order Information: Order ID, Order Date, Shipping Date, Shipping Method, Delivery Days
- Customer and Market Information: Customer Name, Market Segment, Country/Region, Region, Market
- Product Information: Category, Subcategory, Product Name.
- Financial Metrics: Sales, Quantity, Discount, Profit, Shipping Costs, Profit Margin

This dataset has a large sample size, covers multiple regions, and has rich dimensions, making it ideal for analyzing global retail sales trends, profit performance, customer segmentation, and operational efficiency.

2.2 Data Preprocessing

This study applies a standardized data preprocessing pipeline, including data loading, exploratory analysis, data cleaning, type conversion, and feature engineering.

2.2.1 Data Loading & Basic Statistical Analysis

The dataset was loaded using the `pandas.read_csv()` function and encoded using ISO-8859-1 to correctly handle Western European special characters common in international retail data.

Descriptive statistics were generated for the numerical features in the dataset, including five core indicators: sales volume, discount, profit, shipping cost, and year. These statistics provide a comprehensive understanding of the data's central tendency, dispersion, and numerical range. The core statistical results are as follows:

- Quantity: Mean 3.48, ranging from 1 to 14 items, indicating a moderate overall order volume distribution.
- Discount: Mean 0.14, Median 0, Maximum 0.85, indicating that more than half of the orders had no discount.
- Profit: Mean 28.64, Minimum -6599.98, Maximum 8399.98, indicating significant profit fluctuations and the existence of unprofitable orders.
- Shipping_Cost: Mean 26.38, Minimum 0, Maximum 933.57, indicating significant differences in logistics costs among different orders.
- Year: Covering 2011-2014, Mean 2012.78, indicating a uniform temporal distribution of data.

2.2.2 Data Quality Check

Data integrity, uniqueness, and format compliance were verified in batches using code. The core check results are as follows:

- Missing Values: No missing values were detected in any of the 21 features. Each column has 51,290 non-empty records, indicating excellent dataset integrity.
- Duplicate Records: Verification showed zero duplicate rows in the dataset. All transaction records are unique, demonstrating excellent data independence.
- Categorical Feature Validation: The core categorical variables have a clear number of categories, including 4 shipping methods, 3 customer subcategories, 147 countries, 1094 states/provinces, 7 market segments, 13 business regions, 3 product categories, and 17 product subcategories, demonstrating a complete categorization hierarchy.
- Data type check: In the original data, order dates and shipping dates are stored as object (string) types, and the sales amount column, containing commas and currency symbols, is also stored as an object type. This does not meet the requirements for numerical calculations and time analysis, and a specific type conversion is required.

2.2.3 Data Cleaning and Type Conversion

Based on the formatting issues discovered during exploratory data analysis, targeted cleaning

and type standardization were performed on the original data. All original data was retained throughout the process, without deleting any rows or columns, maximizing data integrity:

- **Date Format Standardization:** `order\_date` and `ship\_date` were batch-converted from string types to the standard datetime format. The `dayfirst=True` parameter was set to adapt to internationally accepted date formats, accurately parsing day/month/year format date data and avoiding date recognition errors.
- **Numeric Column Cleaning:** For the non-numerical formatting issues in the sales volume column, commas and currency symbols were removed from the field, and then the data was uniformly converted to floating-point numbers to meet the numerical calculation requirements for subsequent financial calculations and statistical modeling.
- **Data Integrity Protection:** Since the original data had no missing or duplicate values, no data was deleted during the cleaning process, ultimately retaining 51,290 complete transaction records.

2.2.5 Feature Engineering

To support subsequent in-depth retail analysis and uncover more business value, based on the original features, three core derivative features were calculated and extracted to enrich the analysis dimensions:

- **Delivery Time:** A new `delivery\_days` field was added, calculated by the difference between the shipping date and the order date, to accurately measure the logistics and delivery efficiency of each order and reflect the supply chain operation level.
- **Profit Margin:** A new `profit\_margin` field was added, calculated as the ratio of profit to sales revenue, serving as a core financial indicator to objectively assess the profitability of each order and each product category, eliminating profit deviations caused by differences in sales volume.
- **Time Features:** Order month (`order\_month`) and order year (`order\_year`) were extracted from the order date, refining the time dimension to support subsequent annual and monthly sales trend analysis.

2.2.6 Preprocessing Results

After a complete data preprocessing process, the dataset quality and structure were comprehensively optimized. The final processing results are as follows:

- The dataset retains all 51,290 valid transaction records. All field data types were standardized, with no formatting anomalies.
- The valid data period is confirmed to be from January 1, 2011 to December 31, 2014, with a clear time span, suitable for time series analysis.
- All numerical fields such as sales revenue, profit, and shipping costs have been formatted and can be directly used for statistical analysis, visualization, and modeling.
- Four new derived features have been added: delivery days, profit margin, order month, and order year, significantly expanding the analytical dimensions and enhancing the value of data-driven business analysis.

The structured dataset, after cleaning, transformation, and feature derivation, is of high quality, formatted correctly, and has complete dimensions. It provides solid data support for subsequent global retail sales trend analysis, profit optimization, customer behavior analysis, and operational efficiency assessment, ensuring the accuracy and reliability of subsequent analytical conclusions.

3. Multi-Dimensional Data Analysis

This section is divided into four key dimensions. Each team member is responsible for one dimension, including statistical analysis and visualization.

3.1 Dimension 1: Temporal Analysis

This dimension investigates geographic distribution, profit bubbles, and discount sensitivity to identify "profit black holes" and optimize pricing strategies.

- **Global Market Size & Profitability:**

A bubble map analysis of global sales shows that sales are concentrated in a few countries, and not spread across the globe. The US emerges as the top market, both in terms of sales revenue and profit. But the analysis shows several regions are inefficient in terms of profit: several countries have large sales but low profit (bubbles are pale in these regions). This suggests that large markets do not necessarily lead to high profits. We suggest a new multi-faceted strategy that focuses on high-profitability core markets rather than high-revenue markets. In addition, a stringent margin check should be made for big markets with high discounting and shipping costs.

Global Sales & Profit by Country (Bubble = Sales, Color = Profit)

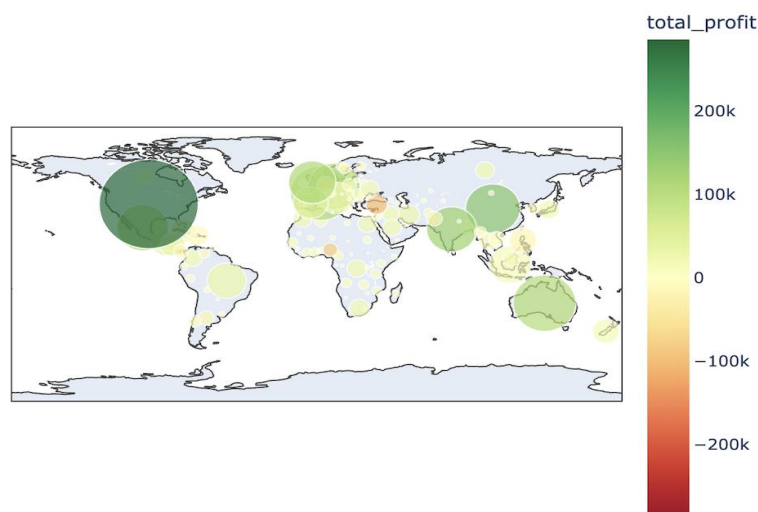


Figure1 Global Sales & Profitability

- **Interactive Global Time Series & Growth Quality:**

We developed an interactive time-series dashboard with dynamic filtering to track sales and profit trends across all global markets (including APAC, Africa, Canada, EU, etc.). This interactive monitoring tool allows us to observe distinct regional growth patterns. While some markets maintain synchronized growth, toggling the dashboard to the APAC market highlights a concerning "sales-profit divergence." During the observed period in APAC, sales grew strongly, but profit growth remained stagnant or negative. This divergence typifies "low-quality growth" driven by low-margin initiatives (like heavy discounting) rather than operational efficiency. We recommend a KPI shift from volume-driven to profit-driven metrics globally, using this interactive tool to dynamically monitor margin health across all regions.

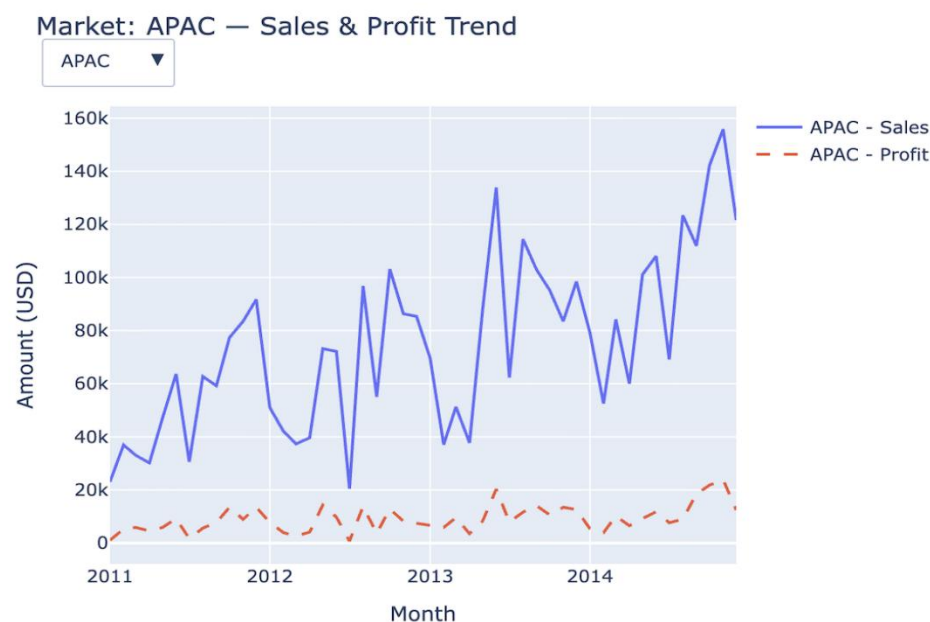


Figure2 Sales & Profit Trend

- **Discount & Margin Erosion:**

A heatmap of the sensitivity of profit margin to discount rates by category reveals key price points. The three product categories all begin with positive profit margins at low discounts, but the margin then declines sharply with higher discounts. The margin break-even point is rigidly located at the 20%-30% discount level; at 30%, the margin becomes negative, destroying value. This problem is common to Office Supplies and Technology categories. Hence, a discount policy should be implemented, with discounts over 30% considered as an exception. Companies should focus on business vitality rather than volume, and tighten discount approvals to avoid lost revenue.



Figure3 Profit Margin Sensitivity to Discount by Category

3.2 Dimension 2: Logistics Efficiency & Probability Distribution (Responsible: Student B)

- Logistics Cost Inversion and Strategic Auditing:

As illustrated in the violin plots in Figure 1, we identified a critical "cost inversion" in the supply chain. Paradoxically, the average cost for Standard Class (\$41.1) is significantly higher than that of First Class (\$20.0), which contradicts traditional service-level pricing. The "long-neck" distribution in Figure 1 highlights extreme outliers in Standard Class, likely driven by unoptimized heavy or cross-border freight.

Business Insight: Based on Figure 1, we recommend a strategic audit of Standard Class orders exceeding \$100. Engaging specialized heavy-freight carriers could eliminate this "high-price, low-efficiency" burden.

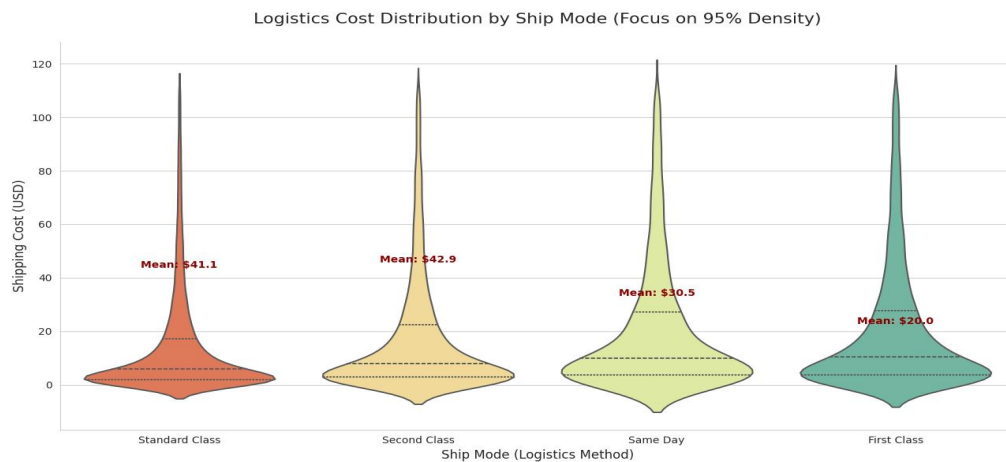


Figure4 Logistics Cost Distribution by Ship Mode

- **The 5-Day "Profit Red Line" and Stability Control :**

As illustrated in the scatter plot in **Figure 2**, there is a decisive correlation between delivery lead times and profit density. By adjusting the **order_priority** slider to "**Medium**" or "**Low**", we observe that **Same Day** shipping (Green dots) acts as a "stability anchor," maintaining concentrated positive margins near the 0-day mark. Conversely, as the delivery window for **Standard Class** (Blue dots) extends from 4 to 7 days, the density of orders falling below the zero-profit line increases significantly. Specifically, the 5-day threshold emerges as a "profit red line," beyond which the probability of loss-making transactions escalates due to potential overheads and inefficiencies.

Strategic Recommendation: To mitigate these risks, we recommend prioritizing the dispatch of Standard Class orders to keep them within the 5-day "safe zone" and leveraging Same Day shipping for high-value clients to ensure margin stability.



Figure5 Impact of Delivery Lead Time on Profitability across Order Priorities

- **Predictive Intelligence for Pre-Order Decision Support :**

To operationalize our findings, we developed the "**Interactive Predictive Dashboard**" shown in **Figure 3**. This system utilizes Logistic and Linear Regression models to transform historical data into proactive guidance. By configuring the parameters—such as setting **Ship Mode** to "First Class" and **Order Priority** to "Critical"—managers can instantly receive a real-time risk assessment. The dashboard identifies the "delay probability" and "estimated profit," providing a clear visual indicator (e.g., Green for low risk, Red for high risk) of the transaction's health.

Strategic Recommendation: This predictive tool enables a shift from "reactive analysis" to "preventative intervention". We advise using this system at the point of sale to flag orders where shipping costs exceed 30% of revenue, allowing for immediate adjustments to pricing or logistics routes.



Figure6 Interactive Predictive Dashboard

3.3 Dimension 3: Product Portfolio Affinity Analysis (Responsible: Student C)

This dimension aims to explore relationships between different products and identify which items are frequently purchased together.

- **Pareto Chart of Sub-Category Profit**

The Pareto Chart shows 80/ 20 distribution in sub-category profitability, in which a small number of products drive the majority of financial performance. The first 8 sub-categories (from copiers to binders) are the major drivers, collectively contributing around 80% of the total profit. In which, copiers and phones are the top two pillars. Copiers and phones contribute 20% and 15% of the cumulative total, respectively.

The remaining 9 categories (from Paper down to Fasteners) contribute only about 20% to the overall bottom line.

We recommend the majority of the marketing budgets, and strict inventory management should be allocated to and imposed on the top 8 categories. Any decline in sales and demand in these areas will have a significant negative impact on the company's financial stability. Meanwhile, since Copiers and Phones are the company's core products, we suggest the company develop defensive strategies (such as loyalty programs or long-term service contracts) to these products to avoid losing the market shares to the competitors.

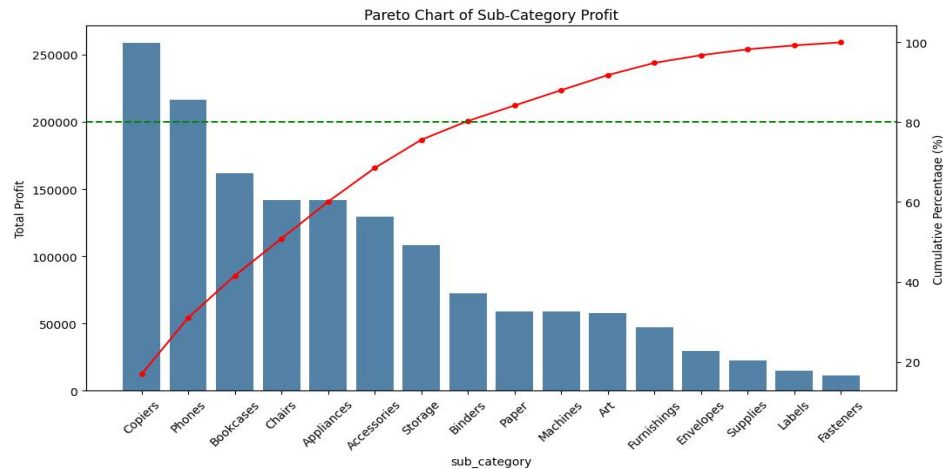


Figure7 Pareto Chart of Sub-Category Profit

- **Sales Size and Profit Margin**

Tables has a relatively large sales volume but negative profit margin (.i.e. deep red). This indicates the more tables sold, the more money lost. Meanwhile, paper and labels have the highest profit margin (.i.e. deep green). Nevertheless, its rectangle size is small, showing it has low sales volume.

Phones are the primary revenue drivers. It has a large area showing high sales volume with healthy profit margins. In the meantime, machines and chairs are in the "pale" zone. They occupy significant space (Sales) but do not generate too much total profit.

We recommend reviewing the cost structure of the Tables via considering raising prices or renegotiating with suppliers for achieving positive margin for the Tables. Meanwhile, we propose launching targeted marketing campaigns for Paper and Labels as they have high profit margins and would have a positive impact on company profit with increasing their sales volume.

We suggest the company maintaining high product standards for phones for keeping the market shares. The company also should investigate how to reduce machines' and chairs' costs to increase their profit margins.

Sales (Size) & Profit Margin (Color)



Figure8 Sales and Profit Margin by Product Category

● Product Association Network

The product association network graph shows appliances, copiers, supplies, chairs, and fasteners are highly interconnected.

The line between envelopes and fasteners, bookcases and copiers, and phones and accessories are short. It means these pairs have strong associations while the customers have common behavior of purchasing them together. In contrast, the line between tables and appliances, and papers and appliances are long, which means the customer's purchase decision logic for tables or paper is relatively independent of appliances.

We recommend the company launch a bundle sale for the highly interconnected products, such as bookcases + copiers + supplies or envelopes + fasteners + supplies to boost the company's total revenue. The company also can create a coupon distributing system, such as if a customer buys a copier but not bookcases, the system should immediately trigger a coupon for bookcases, as data shows they are highly likely to need them.

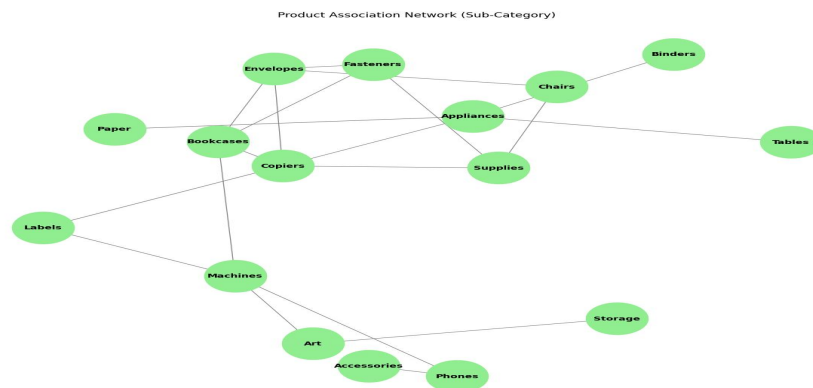


Figure9 Product Sub-Category Association Network

3.4 Dimension 4: Customer Value Tiering & Churn Risk (Responsible: DU Yu)

This dimension analyzes customer behavior through RFM visualization and unsupervised clustering to segment customers, identify high-value groups, and flag churn risks for targeted operational strategies.

- **Customer Value Tiering Visualization & Churn Risk Early Warning**

The radar chart(Figure 10), using standardized RFM metrics, visually validates four distinct customer segments: high-value customers form a full "spider web" with strong performance across recency, frequency, and monetary dimensions; churn-risk customers show low recency scores but solid frequency and monetary values; potential customers exhibit balanced, moderate RFM profiles; and low-value customers have contracted shapes with weak scores across all three metrics.

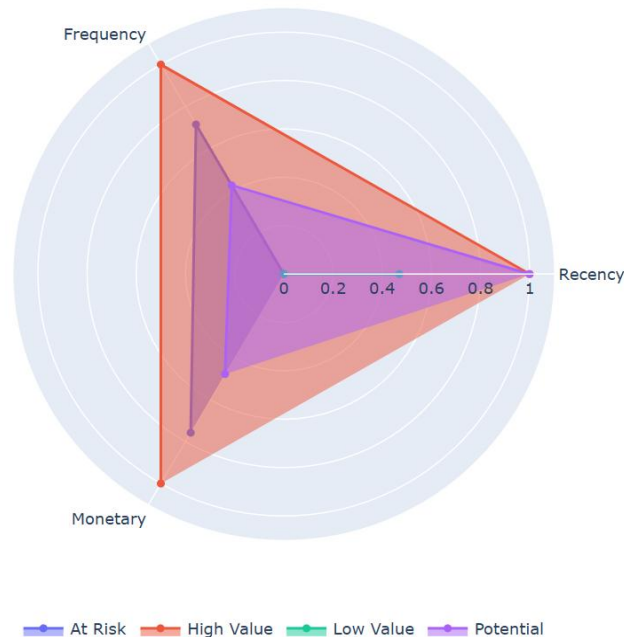


Figure10 Customer Segmentation RFM Radar Chart

Complementing this, the 3D scatter plot maps(Figure 11) individual customers using log-transformed RFM axes, highlighting dense VIP clusters and isolated churn-risk groups, while enabling interactive 360-degree rotation and hover-based detail checks for micro-level insights.

Together, these visualizations form a "profile-position-classify-warn" framework, supporting tiered operations: high-value customers receive exclusive VIP benefits, churn-risk groups get personalized reactivation campaigns, potential customers are targeted with cross-selling to drive value growth, and low-value customers receive low-cost

automated outreach to minimize waste. A monthly monitoring system combining radar chart updates for structural trends and real-time 3D plot tracking for migration, paired with automated alerts for high-frequency/monetary customers with rising recency, shifts churn management from reactive recovery to proactive prevention, while ROI optimization directs 70% of the budget to the top three revenue-contributing segments and reserves 30% for low-value activation testing.

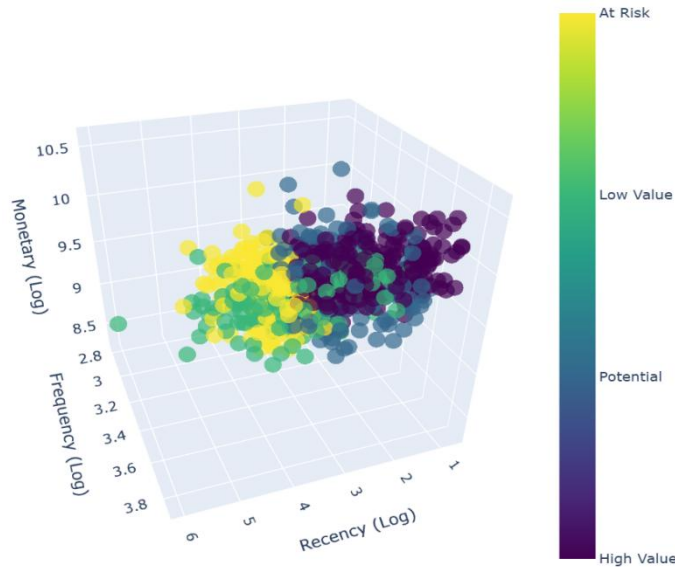


Figure11 3D RFM Customer Segmentation Scatter Plot

- **Customer Value Tiering Visualization & Churn Risk Early Warning**

To refine segmentation, K-Means clustering using the elbow method confirms $k=4$ as optimal, with inertia decreasing rapidly before $k=4$ and slowing afterward(Figure 12).

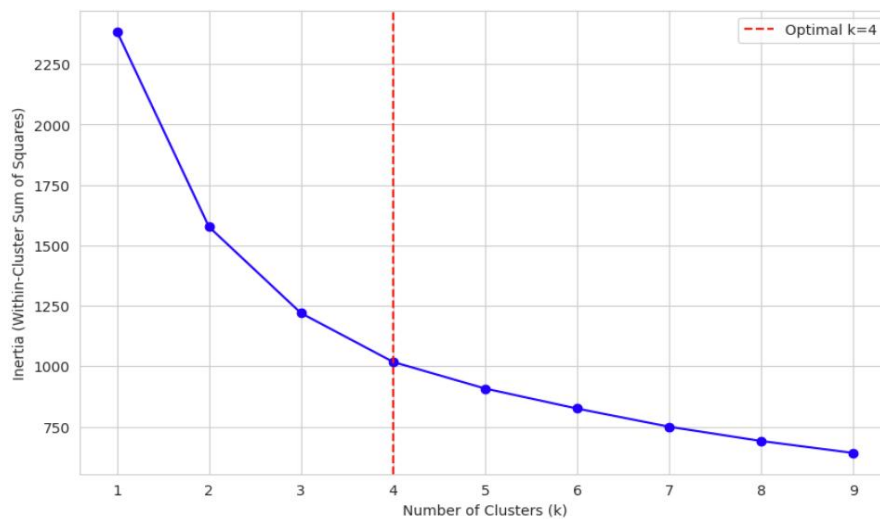


Figure12 Elbow Method for Optimal k (K-Means Clustering)

The four clusters(Figure 13) reveal clear value hierarchies: Cluster 0 (potential customers, 24.8%) shows the lowest recency (4.41 days) and high frequency/monetary values; Cluster 1 (low-value customers, 15.2%) has the lowest frequency and spending; Cluster 2 (high-value customers, 24.3%) leads in frequency and monetary values; and Cluster 3 (at-risk customers, 35.7%) has the highest recency with moderate frequency/monetary values, reflecting dormant high-value behavior.

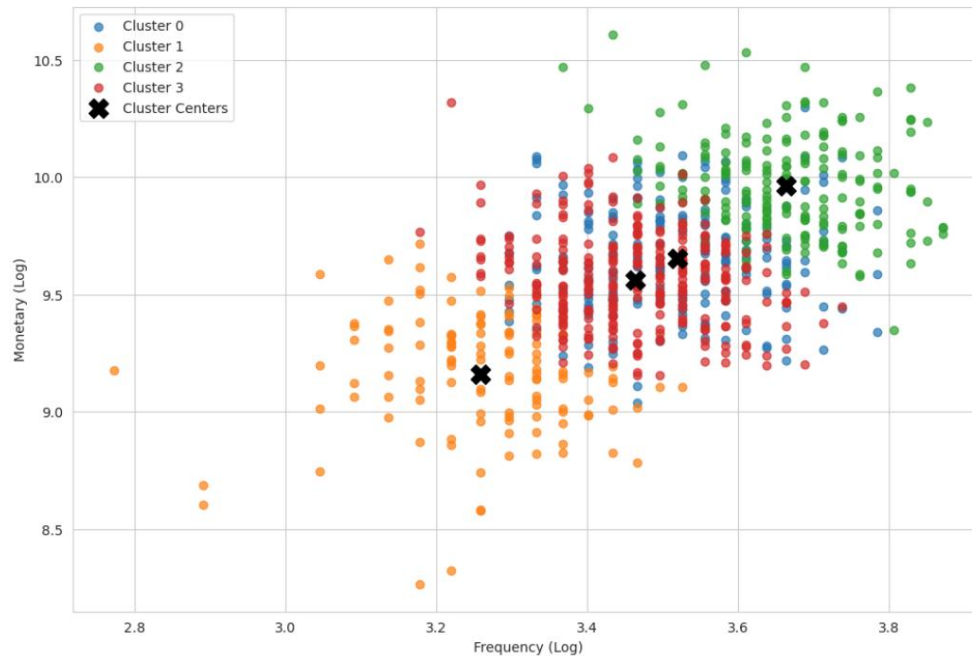


Figure13 K-Means Clustering Results (Frequency vs Monetary)

This distribution aligns with the Pareto principle, where 49% of high-value customers (Clusters 0 and 2) contribute 63% of revenue. Corresponding strategies include loyalty programs for core and potential customers, low-cost outreach for low-value groups, and targeted reactivation campaigns for at-risk customers, supported by monthly clustering reruns to track segment migration and adjust tactics dynamically. Budget allocation mirrors value contribution, directing 70% of resources to high-value, potential, and at-risk customers and using the remaining 30% to test activation for low-value groups, maximizing marketing efficiency and protecting core revenue streams.

4. Integrated Insights & Recommendations

The multi-dimensional analysis reveals that logistics efficiency and customer value are closely linked. High-value customers in specific regions often experience different delivery timelines, which affects their overall satisfaction and churn risk.

Strategic Recommendations:

- **Retention:** Implement a 10% discount strategy for "At Risk" (Cluster 3) customers to prevent churn.
- **Loyalty:** Prioritize VIP maintenance and exclusive offers for "High Value" (Cluster 2)

- customers.
- **Logistics:** Realign shipping protocols in geographic "profit black holes" to reduce costs without sacrificing delivery speed .

5. Conclusion

The project successfully addressed the research question by integrating four distinct analytical dimensions. We identified the primary drivers of profit and the specific customer segments that require strategic attention. While the current models are accurate, future research should incorporate demographic data like age and occupation to enhance customer profiling. This comprehensive approach provides a solid foundation for data-driven decision-making in a global retail environment.

Appendix: Work Allocation

Member Name	Primary Responsibility	Key Deliverables
Zeng Leyao	Dimension 1: Temporal	Time-series analysis and trend visualization.
Chen Zitao	Dimension 2: Logistics	Efficiency auditing and distribution modeling.
Poon Ka Yiu	Dimension 3: Product Portfolio	Affinity analysis and correlation mapping.
Du Yu	Dimension 4: Customer Value	Segmentation, Churn analysis, and RFM modeling.